

Disproof of conjecture 00000001142

earthking11

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Verdict: FALSE

We disprove conjecture 00000001142 as stated. For the mod- p Witt algebra $W(1;1)$ the automorphism group has order 24 at $p = 3$ and 500 at $p = 5$, whereas the conjectured formula $p(p-1)$ gives 6 and 20. The claim is true only at $p = 2$; it fails for *every* $p \geq 3$.

1 The conjecture, quoted verbatim

From `conjectures/00000001142.md`:

English. Conjecture: The order of the automorphism group of the mod- p Witt algebra $W(1;1)$ is $p(p-1)$ (Witt automorphisms).

The conjecture is filed in both English and Chinese; the Chinese original is reproduced verbatim (in Unicode) in `README.md`, and the two versions are identical in content. The PDF font does not embed CJK glyphs, so the Chinese is kept in the markdown copy rather than duplicated here.

2 The algebra and its dimension

Definition 1. $W(1;1) = \text{Der}(\mathbb{F}_p[x]/(x^p))$ is the Lie algebra of derivations of the truncated polynomial ring. It has the p basis elements

$$b_k = x^k \frac{d}{dx}, \quad k = 0, 1, \dots, p-1,$$

and $\dim W(1;1) = p$ (not $p-1$). The bracket is

$$[b_k, b_l] = (l-k) b_{k+l-1} \quad \text{if } 1 \leq k+l \leq p, \quad \text{and } [b_k, b_l] = 0 \text{ otherwise.} \quad (1)$$

For example $b_0 = d/dx$, $b_1 = x d/dx$, ..., $b_{p-1} = x^{p-1} d/dx$.

3 The witness $p = 3$

At $p = 3$ the algebra is 3-dimensional with basis b_0, b_1, b_2 and bracket table

$$[b_0, b_1] = b_0, \quad [b_0, b_2] = 2b_1, \quad [b_1, b_2] = b_2,$$

the other brackets following from antisymmetry ($[b_k, b_k] = 0$, $[b_l, b_k] = -[b_k, b_l]$).

A linear map is a 3×3 matrix over $\mathbb{F}_3$; we enumerated *all* $3^9 = 19683$ of them:

candidate matrices	$3^9 = 19683$
invertible ($\det \neq 0$)	$ \mathrm{GL}(3, \mathbb{F}_3) = 11232$
invertible <i>and</i> bracket-preserving	24
conjectured value $p(p-1) = 3 \cdot 2$	6

Hence

$$|\mathrm{Aut}(W(1;1))| = 24 \neq 6 = p(p-1) \quad (p=3)$$

and the conjecture is false at $p = 3$. The count 24 is machine-verified twice: by the exhaustive enumeration in `reproduce.py` and by the core-Lean theorem `aut_count_3` in `lean4/Main.lean`.

Structural confirmation: $W(1;1) \cong \mathfrak{sl}(2, \mathbb{F}_3)$

At $p = 3$ the algebra is isomorphic to $\mathfrak{sl}(2, \mathbb{F}_3)$ via

$$h = 2b_1, \quad e = b_2, \quad f = 2b_0,$$

for which one checks $[h, e] = 2e$, $[h, f] = -2f$, $[e, f] = h$. Consequently

$$|\mathrm{Aut}(W(1;1))| = |\mathrm{Aut}(\mathfrak{sl}(2, \mathbb{F}_3))| = |\mathrm{PGL}(2, 3)| = |S_4| = 24.$$

This is an independent theoretical confirmation of the numerical value 24. The element-order distribution of the 24 automorphisms is

$$\{1 : 1, 2 : 9, 3 : 8, 4 : 6\},$$

which is exactly the order distribution of S_4 ; it is computed in `reproduce.py`. Note that the exceptional isomorphism with $\mathfrak{sl}(2, \mathbb{F}_3)$ exists only at $p = 3$.

4 The value at $p = 5$

At $p = 5$ the algebra is 5-dimensional. A direct brute force over 5^{25} matrices is infeasible, so we use the following exact structural reduction: the pair $\{b_0, b_4\}$ generates $W(1;1)$ as a Lie algebra, so an automorphism φ is determined by the pair $(\varphi(b_0), \varphi(b_4))$; and $\varphi(b_0)$ must be a *regular nilpotent* element, because $\mathrm{ad}(\varphi(b_0)) = \varphi \circ \mathrm{ad}(b_0) \circ \varphi^{-1}$ is conjugate to $\mathrm{ad}(b_0)$, a single Jordan block of size p . There are exactly 500 regular nilpotent elements. Enumerating this pair for all of them, reconstructing $\varphi(b_3), \varphi(b_2), \varphi(b_1)$ from the relations $[b_0, b_k] = k b_{k-1}$ and testing bracket preservation and invertibility gives

regular nilpotent candidates for $\varphi(b_0)$	500
extensions to an automorphism (each candidate has exactly one)	500
conjectured value $p(p-1) = 5 \cdot 4$	20

Hence

$$|\mathrm{Aut}(W(1;1))| = 500 = (p-1)p^{p-2}|_{p=5} \neq 20 = p(p-1) \quad (p=5)$$

and the conjecture is false at $p = 5$ as well. This computation is carried out exactly in `reproduce.py`; it was validated against the $p = 3$ brute force (where it returns 24).

5 The exception $p = 2$

At $p = 2$ the algebra is the 2-dimensional non-abelian Lie algebra $[b_0, b_1] = b_0$ (since $k + l = 1 \leq 2$), and $|\text{Aut}| = 2 = p(p - 1)$. So the conjectured formula is correct at $p = 2$ and wrong for every $p \geq 3$; the number $p(p - 1)$ is therefore not the order of $\text{Aut}(W(1; 1))$ in general.

6 Rescue attempts that fail

We tried to save the conjecture by changing the reading of “Aut”; none succeeds.

Reading	Outcome
Automorphisms of the <i>restricted</i> Lie algebra (respecting the p -power map $X \mapsto X^{[p]}$)	the same counts result: at $p = 3$ all 24 Lie automorphisms preserve the p -power map, so the restricted automorphism group is again 24; likewise the count at $p = 5$ is again 500. No rescue.
A different (non-standard) p -power convention	gives 2 at $p = 3$ and 4 at $p = 5$, even further from $p(p - 1)$. No rescue.
Inner automorphisms	$W(1; 1)$ is finite-dimensional, so the inner automorphism group is trivial and $\text{Out} = \text{Aut}$, with order 24 (resp. 500). No rescue.
Automorphisms of the restricted enveloping algebra	a different (larger, associative-algebra) group; not $\text{Aut}(W(1; 1))$. No rescue.

7 The Möbius subgroup: where $p(p - 1)$ really comes from

The number $p(p - 1)$ does occur, but as the order of a *proper* subgroup. The Möbius/Witt transformations

$$x \mapsto \frac{ax}{1 + bx}, \quad a \in \mathbb{F}_p^\times, \quad b \in \mathbb{F}_p,$$

act on vector fields by pullback, $\varphi^*(b_k) = \varphi(x)^k / \varphi'(x) \cdot d/dx$, i.e.

$$\varphi^*(b_k) = a^{k-1} x^k (1 + bx)^{2-k} \frac{d}{dx},$$

expanded as a power series in x and truncated at degree $p - 1$. This action preserves the bracket (1), and the $p(p - 1)$ maps (a, b) form a subgroup of $\text{Aut}(W(1; 1))$:

$$|\{x \mapsto ax/(1 + bx)\}| = p(p - 1) = 6 \quad (p = 3), \quad 20 \quad (p = 5).$$

But these are *proper* subgroups, since

$$6 < 24 = |\text{Aut}| \quad (p = 3), \quad 20 < 500 = |\text{Aut}| \quad (p = 5).$$

Thus the conjecture is true exactly if “Aut” is silently read as this Möbius/Witt subgroup rather than the full automorphism group. Under the literal reading of the conjecture as filed, it is false for all $p \geq 3$.

8 Reproducibility

```
$ python3 reproduce.py
... (brute force at  $p=3$ ; the exact  $p=5$  count; the Möbius subgroup; the
 $p=2$  case)
PASS: all 21 checks verified; conjecture 00000001142 is FALSE.
```

The script uses only the Python 3 standard library. It prints the $|\mathrm{GL}(3, \mathbb{F}_3)| = 11232 / 24$ breakdown so that the enumeration is independently checkable, and it exits 0 exactly when every check holds. Runtime is about 13 seconds.

9 Formalisation

The $p = 3$ refutation is formalised in the Lean 4 project under `lean4/` (core Lean only, `import Std`, no `Mathlib`, no `sorry`). The main statements are

```
theorem num_mats : allMats.length = 3 ^ 9
theorem aut_count_3 : auts3.length = 24
theorem claim_false_3 : auts3.length ≠ 3 * (3 - 1)
```

The algebra is defined by its nine structure constants only, and the number 24 is produced by exhaustively filtering the 3^9 matrices over $\mathbb{F}_3$. The enumeration is discharged by the core kernel `decide` tactic, *not* by `native_decide`; the axiom audit (`lake env lean Check.lean`) reports that the theorems depend on no axioms at all. The measured `lake build` wall time is about 50 seconds (47.5 s on the final clean build). The case $p = 5$ is out of scope for Lean (a faithful analogue would need 5^{25} matrices) and is covered by `reproduce.py`. See `lean4/README.md` for details.

Status against the submission rules

Rule 3 requires a LaTeX source, a PDF document, and a Lean 4 project.

- **LaTeX source** — `main.tex`, a standalone article using `geometry`, `amsmath`, `amssymb`, `amsthm`, `array`, `parskip`, `booktabs`; compiles with `tectonic --outdir build main.tex`.
- **PDF** — at `build/main.pdf`.
- **Lean 4 project** — `lean4/` with `lean-toolchain` pinned to `leanprover/lean4:v4.33.1`, `lakefile.toml` (name `tlmc1142`, library `Main`, no dependencies), `Main.lean`, `Check.lean` (`#print axioms`) and `lean4/README.md`. `lake build` exits 0 in about 50 seconds and the audit reports no `sorryAx` and no `Lean.ofReduceBool`.